Olympiad Test Paper — Social Science (Class 7) — Paper 3

Chapter 2: Understanding the Weather

Paper 3 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	2 — Understanding the Weather
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch for confusing weather with climate, assuming pressure rises as you climb (it falls), swapping the jobs of the thermometer/barometer/rain gauge/wind vane/anemometer/hygrometer, and treating high humidity as a guarantee of rain. Do not write on this paper — mark answers on your answer sheet.

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1. Reena writes two notes about Shimla. Note 1: "This afternoon it is 9 °C, cloudy and breezy." Note 2: "Shimla is cool in summer and snowy in winter, year after year." Which note describes the WEATHER and which the CLIMATE?

(A) Note 1 is climate and Note 2 is weather
(B) Note 1 is weather (the state at a particular time) and Note 2 is climate (the long-term average pattern)
(C) Both notes describe weather because both mention Shimla
(D) Both notes describe climate because both mention temperature

2. A traveller says, "I packed for Jaisalmer expecting a hot, dry desert, but on the day I arrived it was actually cool and raining." This situation shows that:

(A) The climate of Jaisalmer must have permanently changed to wet
(B) Weather and climate are the same thing, so the traveller was simply wrong
(C) Rain is impossible in any place that has a desert climate
(D) Climate tells the usual long-term pattern, but the weather on any single day can differ from it

3. Most clouds, rain and storms form in the troposphere, the lowest atmospheric layer. The troposphere is thicker over the tropics than over the poles. The most consistent reason within the chapter is that:

(A) Cold polar air expands more than warm tropical air

(B) The poles receive more rainfall, which thickens the layer

(C) The tropics are physically much closer to the Sun than the poles are

(D) Warm tropical air expands and rises, pushing the layer higher than the cold, shrunken air at the poles does

4. A weather club is sorting cards into "element of weather" and "not an element of weather." Which set of FOUR cards belongs entirely in the "element of weather" pile?

(A) Temperature, soil type, rainfall, wind(B) Humidity, rivers, pressure, sunshine

(C) Wind, mountains, temperature, precipitation

(D) Temperature, atmospheric pressure, humidity, wind

5. A pilot needs to know the conditions of the air at one airport at one moment so a plane can take off safely. The pilot is asking about:

(A) The weather, because it is the current state of the atmosphere at that place

(B) The climate, because aircraft depend on long-term averages

(C) The climate, because airports record thirty-year patterns

(D) Neither weather nor climate, because air conditions cannot affect flight

6. Two friends describe the same morning. One says "it feels a bit chilly," the other says "the thermometer reads 12 °C." Why is the second statement more useful to a weather station?

(A) It makes the morning warmer than "chilly" does

(B) "Chilly" can only ever describe wind, never temperature

(C) A spoken feeling is always more accurate than an instrument

(D) It gives an exact, shared value on a fixed scale that anyone can compare

7. A student lists the elements that must be measured to describe weather but slips in one wrong entry: temperature, pressure, humidity, wind, precipitation, soil fertility. Which entry must be removed, and why?

(A) Soil fertility, because it is a property of the land, not a condition of the atmosphere

(B) Pressure, because it cannot be measured with an instrument

(C) Humidity, because it is the same thing as precipitation

(D) Wind, because wind is part of the climate but not the weather

8. Match each element to its correct unit. The one CORRECT pairing of element and unit is:

(A) Temperature → millibars (mb) (B) Atmospheric pressure → millibars (mb)

(C) Rainfall → percent (%) (D) Humidity → degrees Celsius (°C)

9. In a liquid-in-glass thermometer the coloured liquid climbs the thin tube as the day warms. The chain of cause and effect is:

(A) The tube widens in the heat, letting the liquid pour upward

(B) Warm air is pushed down into the tube, lifting the liquid

(C) The liquid is heated, it expands, and the expanded liquid is forced up the narrow tube

(D) Heat makes the liquid lighter so it simply floats out of the bulb

10. A weather log gives a temperature first as 20 °C and on a different day as 68 °F. A student panics that the thermometer is faulty because the numbers are so different. The correct reading of this is:

(A) 68 °F must be far hotter than 20 °C because 68 is the bigger number

(B) Fahrenheit actually measures pressure, so the two cannot be compared

(C) Celsius and Fahrenheit are two different scales, so the same warmth shows different numbers on each—nothing is faulty

(D) The thermometer is broken because one temperature should give one number only

11. A station's record for one day: maximum 34 °C at 3 p.m., minimum 22 °C at dawn. The daily RANGE of temperature is:

(A) 28 °C

(B) 56 °C

(C) 12 °C

(D) 22 °C

12. Using the same day (maximum 34 °C, minimum 22 °C), the MEAN (average) daily temperature is:

(A) 12 °C

(B) 28 °C

(C) 56 °C

(D) 34 °C

13. Four towns reported the same day. Which town had the LARGEST daily temperature range?

Town | Max (°C) | Min (°C)

P | 30 | 24

Q | 41 | 23

R | 28 | 19

S | 35 | 30

(A) Town Q, with a range of 18 °C

(B) Town P, with a range of 6 °C

(C) Town S, because it has the second-highest maximum

(D) Town R, because it has the lowest minimum of the four

14. Reading the same table, which town stayed most STEADY in temperature across the day (smallest range)?

Town | Max (°C) | Min (°C)

P | 30 | 24

Q | 41 | 23

R | 28 | 19

S | 35 | 30

(A) Town R, because it has the lowest maximum
(B) Town Q, because its numbers are the largest
(C) Town S, with a range of only 5 °C (D) Town P, with a range of 12 °C

15. Town A's daily range is 4 °C and town B's is 17 °C on the same day. The MOST that can be concluded is:

(A) Town A was definitely hotter than town B all day
(B) Town B definitely received more rainfall than town A
(C) Town B's temperature swung much more between its daytime high and night-time low than town A's did
(D) Town A's average temperature must be lower than town B's

16. Snow falls into an open rain gauge overnight and is still solid in the morning. To get a correct reading the observer should:

(A) Let the snow melt to water and then measure the depth of that water on the scale
(B) Read the depth of the solid snow directly as the rainfall figure
(C) Tip the snow out and record the rainfall as zero
(D) Double the snow depth to allow for melting

17. A rain gauge reading of "15 mm" for the day means that:

(A) Exactly 15 millilitres of water were collected in total
(B) It rained for exactly 15 minutes
(C) If the rain had stayed where it fell on flat ground, it would lie 15 mm deep
(D) The temperature fell by 15 degrees during the rain

18. Hail and rain are both forms of precipitation. The key difference between them is that:

(A) Hail is water vapour in the air, while rain is liquid water
(B) Hail falls only in summer, while rain falls only in winter
(C) Hail is measured in percent, while rain is measured in degrees
(D) Hail reaches the ground as hard balls of ice, while rain reaches it as liquid water droplets

19. Two gauges sit side by side. Gauge 1 reads 4 mm and Gauge 2 reads 40 mm after the same storm passes the city. The most sensible thought is:

(A) The storm dropped exactly ten times more water on one gauge for no reason
(B) Gauge 2 must be measuring temperature instead of rain

(C) Rainfall depth always doubles every few metres across a city

(D) One gauge is probably blocked or faulty, since the same storm over one small area should give similar readings

20. Atmospheric pressure is best understood as:

(A) The push of the weight of all the air stacked above a place, pressing down on it

(B) The amount of invisible water vapour in the air

(C) The speed at which the wind is moving across the ground

(D) The temperature of the air high in the sky

21. A mountaineer climbs from a coastal town up to a high pass. As she gains height, the atmospheric pressure she experiences:

(A) Falls, because less and less air remains above her to press down

(B) Rises, because she is getting nearer to the top of the sky

(C) Stays exactly the same, since air is everywhere

(D) Jumps suddenly to zero the moment she reaches the pass

22. A hill station reads 700 mb while a coastal town reads 1010 mb at the very same hour. The best explanation is:

(A) The hill station's barometer must be broken

(B) The coast is much colder than the hill station that day

(C) The hill station was simply having heavier rainfall

(D) The hill station is much higher up, so there is less air above it pressing down

23. A child argues, "On a tall mountain you are closer to the top of the sky, so the pressure there must be higher." The flaw is that pressure depends on:

(A) How close you are to the Sun

(B) The temperature of the rock you are standing on

(C) How recently it last rained at that spot

(D) The weight of the air ABOVE you, and there is less of it the higher you climb

24. Normal sea-level pressure is about 1013 mb. A region where the barometer drops well below this, say to 990 mb, is called a depression. Its importance is that it:

(A) Always brings clear, calm and settled weather

(B) Means the barometer has stopped working

(C) Can develop into a storm, so falling pressure is an early warning sign

(D) Shows the air has become drier and cooler with no risk

25. A barometer needle has been steadily climbing for several hours, settling toward high pressure. The most likely change in the weather is:

(A) Calmer, clearer, more settled weather is on the way

(B) A storm is about to break, because rising pressure means storms

(C) Heavy rain will start immediately

(D) The wind will stop completely and the air will become wet

26. Wind is air in motion. Wind blows because air moves:

(A) From places of lower pressure toward places of higher pressure

(B) From colder places to warmer places, regardless of pressure

(C) From places of higher pressure toward places of lower pressure

(D) From wetter places to drier places, regardless of pressure

27. Region M is at 1020 mb and region N, right beside it, is at 1004 mb. The wind between them will:

(A) Blow from M toward N, that is from the higher pressure to the lower

(B) Blow from N toward M, from lower pressure to higher

(C) Spin in a closed circle and go nowhere

(D) Rise straight upward and not move sideways at all

28. Two pairs of regions each have a pressure difference. Pair 1 differs by 4 mb; Pair 2 differs by 20 mb, over the same distance. Compared with Pair 1, the wind in Pair 2 will most likely be:

(A) Weaker, because a larger difference slows the air down

(B) Stronger, because a larger pressure difference drives a faster wind

(C) Exactly the same speed, since wind speed ignores pressure

(D) Reversed in direction compared with Pair 1

29. A weather station must record BOTH the direction the wind comes from AND its speed. The correct pair of instruments is:

(A) A barometer for direction and a thermometer for speed

(B) An anemometer for direction and a wind vane for speed

(C) A rain gauge for direction and a hygrometer for speed

(D) A wind vane for direction and an anemometer for speed

30. When the wind blows, a wind vane's tail is pushed downwind so its arrow points into the wind. This lets the vane report:

(A) How fast the wind is blowing

(B) The direction from which the wind is coming

(C) How much water vapour the air holds

(D) How many raindrops are falling each minute

31. An anemometer's cups are spinning very fast. The reading this gives is:

(A) A high atmospheric pressure (B) A high relative humidity

(C) The exact direction the wind is coming from

(D) A high wind speed

32. At an airport a fabric wind sock streams out nearly straight and level. A pilot reads from it that the wind is:

(A) Almost calm, since a level sock means little wind

(B) Bringing heavy rain to the runway (C) Raising the air pressure over the field

(D) Blowing fairly strongly, and the sock also shows its direction

33. Relative humidity is the amount of water vapour in the air compared with the most it could hold, given as a percentage. A reading of 95 % therefore means the air is:

(A) Almost completely dry

(B) Almost as full of water vapour as it can hold—very damp

(C) Definitely raining at that very moment(D) At a temperature of about 95 degrees

34. Dry weather typically shows relative humidity around 20–40 % and humid weather around 60–80 %. A station that reads 30 % is best described as:

(A) Damp, sticky air (B) Dry, crisp air

(C) Air in which it must be raining

(D) Air that is completely saturated with water vapour

35. Two cities on the same warm day: Kochi reads 85 % humidity, Jodhpur reads 25 %. Wet washing hung outdoors will dry FASTER in Jodhpur because there:

(A) The air is far from full of vapour, so it can readily absorb more moisture from the clothes

(B) The air is too cold to let water evaporate

(C) The high pressure presses the water into the cloth

(D) The wind is forced to stop over the clothes

36. On a muggy day a hygrometer reads 90 %, yet no rain falls all afternoon. The best explanation is:

(A) High humidity always brings rain, so the hygrometer must be wrong

(B) Humidity and rainfall are completely unrelated to each other

(C) The reading of 90 % actually means the air is very dry

(D) High humidity alone is not enough; the air must cool enough for the vapour to condense before rain forms

37. A student claims, "A barometer and a thermometer measure pretty much the same thing." The correct reply is:

(A) No—the barometer measures atmospheric pressure while the thermometer measures temperature, two different elements

(B) Yes, both of them measure temperature(C) Yes, both of them measure humidity

(D) No—the barometer measures rainfall and the thermometer measures wind

38. Which one of these element-to-instrument pairings is INCORRECT?

(A) Humidity → barometer
(B) Temperature → thermometer
(C) Rainfall → rain gauge
(D) Wind speed → anemometer

39. A weather station owns a thermometer, a rain gauge and a hygrometer but no barometer. Which weather element can it NOT measure?

(A) Temperature
(B) Atmospheric pressure
(C) Rainfall
(D) Humidity

40. An observer wants to know which way the wind comes from and also how moist the air is. The correct pair of instruments is:

(A) An anemometer and a barometer
(B) A rain gauge and a thermometer
(C) A wind vane and a hygrometer
(D) A barometer and a wind vane

41. Before instruments existed, people often forecast rain by watching nature—for example, ants carrying their eggs to higher ground or birds flying low. A real WEAKNESS of relying only on such clues is that they:

(A) Are always perfectly accurate and never wrong
(B) Give exact numbers for pressure and humidity
(C) Are tied to one spot and give only vague, short-range hints like "rain is coming"
(D) Cover the whole country with one reading

42. Meteorology differs from simply reading nature's clues mainly because it:

(A) Measures each weather element precisely with instruments and gathers data from many places over time
(B) Relies on watching a single animal as its only guide
(C) Refuses to use any instruments at all
(D) Can describe only today and can never predict ahead

43. An Automated Weather Station (AWS) is especially valued in remote, dangerous places because it:

(A) Needs an operator to read each instrument every hour
(B) Uses sensors to record data on its own, with no person needing to stay there
(C) Works only inside large cities during the daytime
(D) Measures the region's climate instead of its weather

44. Which body in India gathers weather data nationwide and issues forecasts and warnings such as cyclone alerts?

(A) The National Rain Gauge Office
(B) The Department of Climate History
(C) The India Meteorological Department
(D) The Indian Wind Vane Society

45. Day and night, and the whole pattern of weather, are ultimately driven by:

(A) The heat of the Sun, which warms the air and surfaces unevenly
(B) The Moon's gravity pulling on the air

(C) The spinning of the wind vanes at weather stations

(D) The barometers raising and lowering the pressure

46. A bare patch of ground gets hotter than a nearby pond under the same midday Sun. The air just above the ground rises while air over the pond stays heavier and sinks. This shows that the Sun causes wind by:

(A) Heating everything exactly equally everywhere

(B) Directly blowing the air with its rays

(C) Heating surfaces unevenly, which sets up pressure differences that make air flow

(D) Cooling the ground so the air freezes in place

47. Why are the tropics generally warmer than the polar regions, even though both are roughly the same distance from the Sun?

(A) The tropics are physically much closer to the Sun than the poles

(B) The poles receive sunlight for far more hours every single day all year

(C) Cold air from space pours only onto the tropics

(D) Sunlight strikes the tropics more directly, so its heat is concentrated, while it strikes the poles at a slant and spreads out

48. After the Sun sets, the land cools and the temperature usually falls to its lowest just before dawn. This daily rise-and-fall of temperature is direct evidence that:

(A) The barometer controls the temperature each day

(B) Humidity is what makes the Sun rise and set

(C) Wind speed decides whether it is day or night

(D) The Sun is the main source of heat that drives the day's weather changes

49. A weather log for one afternoon:

Time | Pressure (mb) | Humidity (%) | Wind | Sky

1 p.m. | 1010 | 55 | light | sunny

2 p.m. | 1004 | 70 | rising | cloud building

3 p.m. | 998 | 85 | strong | dark cloud

The best forecast for the evening is:

(A) Clear, calm and dry weather

(B) Wet, stormy weather is on the way

(C) A cold, cloudless evening

(D) No change at all from the 1 p.m. conditions

50. In that same log, the steadily FALLING pressure (1010 → 1004 → 998 mb) is an early warning because falling pressure often marks:

(A) The arrival of clear, settled fair weather

(B) The approach of a depression, the seed of a storm

(C) A rise in the station's height above the sea

(D) The complete end of all wind in the area

51. A station's three readings are mixed up: 1008, 27, 65. They belong to pressure (mb), temperature (°C) and humidity (%). The most sensible assignment is:

(A) 1008 °C temperature, 27 % humidity, 65 mb pressure

(B) 1008 % humidity, 27 mb pressure, 65 °C temperature

(C) 1008 mb pressure, 27 °C temperature, 65 % humidity

(D) 27 mb pressure, 65 °C temperature, 1008 % humidity

52. A forecast warns fishermen of a coming storm so they stay ashore, and warns a city of likely flooding so drains are cleared. This shows the chief value of forecasts is that they:

(A) Stop the storm or flood from ever forming

(B) Give people and authorities time to prepare and stay safe before severe weather arrives

(C) Make the weather change less often than before

(D) Remove the need for any weather instruments

53. Bringing a thermometer, rain gauge, barometer, wind vane, anemometer and hygrometer together in one place and reading them at fixed times is the purpose of a:

(A) Climate map (B) Wind sock

(C) Weather station (D) Depression

54. A traveller reaching a high mountain pass feels breathless and light-headed. Linking pressure and altitude from the chapter, the best reason is that high up the air is:

(A) Heavier, with more oxygen packed into it than at the coast

(B) Thinner, with lower pressure and therefore less oxygen in each breath

(C) Completely empty of all gases at that height

(D) So full of rain that it blocks normal breathing

55. Assertion (A): Atmospheric pressure decreases as one climbs a mountain.

Reason (R): Higher up, there is less air left above to press down.

Choose the best option:

(A) Both A and R are true, and R correctly explains A

(B) Both A and R are true, but R does not explain A

(C) A is true but R is false (D) A is false but R is true

56. Assertion (A): High relative humidity always means it is raining.

Reason (R): Rain needs the air to cool enough for water vapour to condense.

Choose the best option:

(A) A is false but R is true (B) Both A and R are true, and R explains A
(C) Both A and R are true, but R does not explain A
(D) A is true but R is false

57. Assertion (A): A wind vane and an anemometer measure the same property of the wind.

Reason (R): A wind vane shows wind direction while an anemometer measures wind speed.

Choose the best option:

(A) Both A and R are true, and R explains A(B) A is false but R is true
(C) Both A and R are true, but R does not explain A
(D) A is true but R is false

58. Assertion (A): Wind blows from a low-pressure area toward a high-pressure area.

Reason (R): Air always moves to balance out differences in pressure.

Choose the best option:

(A) Both A and R are true, and R explains A(B) A is false but R is true
(C) Both A and R are true, but R does not explain A
(D) A is true but R is false

59. A class concludes: "Studying Mumbai's weather for one rainy week tells us Mumbai's climate." The flaw in this reasoning is that:
(A) One week is more than enough to fix a place's climate forever
(B) Weather and climate mean exactly the same thing anyway
(C) A rainy week proves Mumbai's climate is permanently dry
(D) Climate is the average over many years, so one week of weather cannot define it

60. Putting the chapter together, which statement is COMPLETELY correct?
(A) Pressure rises with altitude, is measured by a thermometer, and a drop means clear skies
(B) Humidity is measured by an anemometer in degrees, and 100 % means very dry air
(C) Pressure falls with altitude, is measured by a barometer in millibars, and a sharp drop can warn of a storm
(D) Rainfall is measured by a wind vane in percent, and high humidity always brings rain

Solutions — Social Science (Class 7), Chapter 2: Understanding the Weather — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	A	31	D	41	C	51	C
2	D	12	B	22	D	32	D	42	A	52	B
3	D	13	A	23	D	33	B	43	B	53	C
4	D	14	C	24	C	34	B	44	C	54	B
5	A	15	C	25	A	35	A	45	A	55	A
6	D	16	A	26	C	36	D	46	C	56	A
7	A	17	C	27	A	37	A	47	D	57	B
8	B	18	D	28	B	38	A	48	D	58	B
9	C	19	D	29	D	39	B	49	B	59	D
10	C	20	A	30	B	40	C	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Weather is the condition of the atmosphere at a given moment and place, so a single cloudy 9 °C afternoon is weather. Climate is the average atmospheric pattern of a place over many years, so the repeated cool-summer/snowy-winter description is climate. Mentioning the same place or mentioning temperature does not decide between the two; what matters is whether the statement is about one moment or about a long-term, year-after-year average.

2. (D) Climate is the average pattern expected over years, while weather is what actually happens on a particular day, and a single day can depart from the average. One cool rainy day does not redefine a region's climate, nor does it prove weather and climate are identical, and even dry-climate regions get occasional rain.

3. (D) Warm air expands and takes up more vertical space, so the weather layer reaches higher over the warm tropics; cold polar air contracts, keeping the layer thinner. Cold air does not expand more than warm air, rainfall amount does not set the layer's thickness, and all parts of Earth are at essentially the same distance from the Sun—the tropics are warmer because sunlight strikes them more directly, not because they are closer.

4. (D) Elements of weather describe the changing state of the air: temperature, atmospheric pressure, humidity, wind, precipitation and sunshine/cloud cover. Soil type, rivers and mountains are features of the land, not conditions of the atmosphere, so any set containing them is mixed and incorrect.

5. (A) A take-off decision depends on conditions right now—wind, visibility, rain—so the pilot needs the weather, the present state of the atmosphere. Climate, the multi-decade average, cannot tell whether it is safe to take off in the next few minutes, and air conditions clearly do affect flight.

6. (D) Personal words like "chilly" depend on who is speaking and cannot be compared between places, whereas a measured 12 °C is a fixed number on a shared scale that everyone reads the same way. The reading does not change the actual temperature, "chilly" does refer to how cold it feels, and instruments are more reliable than feelings for sharing data.

7. (A) Weather elements describe the state of the air, and soil fertility is a quality of the ground, so it does not belong. Pressure is measurable with a barometer, humidity (water vapour in the air) is different from precipitation (water that falls), and wind is very much a weather element.

8. (B) Atmospheric pressure is reported in millibars. Temperature is in degrees (°C or °F), rainfall depth is in millimetres, and relative humidity is in percent—so each of the other pairings swaps the unit onto the wrong element.

9. (C) Heating makes the liquid expand; because the only room to grow is up the narrow bore, the level rises and we read a higher temperature. The glass tube does not noticeably widen, no outside air enters the sealed tube, and the rise is caused by expansion taking up space, not by the liquid "floating."

10. (C) Temperature can be reported on more than one scale, and 20 °C is in fact the same warmth as 68 °F, so a larger Fahrenheit number does not mean a hotter day. Fahrenheit measures temperature, not pressure, and a single temperature legitimately has different values on different scales—so the instrument is not at fault.

11. (C) The daily range is the highest reading minus the lowest, so 34 − 22 = 12 °C. Adding and halving (28) gives the mean, not the range; adding the two readings (56) is meaningless here; and 22 °C is merely the minimum, not the spread.

12. (B) The mean daily temperature is the average of the maximum and minimum: (34 + 22) ÷ 2 = 28 °C. 12 °C is the range (the difference), 56 °C is the un-halved sum, and 34 °C is only the maximum.

13. (A) The range is max minus min for each town: P = 6, Q = 18, R = 9, S = 5. The largest difference, 18 °C, belongs to Q. A high maximum (S) or a low minimum alone does not decide the range—only the gap between a town's own high and low does.

14. (C) Ranges are P = 6, Q = 18, R = 9, S = 5, so S varied least. A low maximum (R) does not by itself mean a small swing, large numbers (Q) actually show the biggest swing here, and P's range is 6, not 12.

15. (C) Range measures only the gap between a place's own maximum and minimum, so a larger range means a bigger day-to-night swing—nothing more. It says nothing about which town was hotter, which got more rain, or which had the higher average, since two places can share an average yet have very different ranges.

16. (A) A rain gauge reports the depth of liquid water, so frozen precipitation must be melted first and the resulting water measured. Reading solid snow depth overstates the figure (snow is far less dense than water), recording zero ignores real precipitation, and doubling is an invented rule with no basis.

17. (C) Rainfall is measured as a depth: 15 mm is how deep the water would stand on level ground that neither soaked it in nor let it run off. Millilitres are a volume, not a depth; the figure says nothing about how long it rained or about temperature change.

18. (D) Precipitation is water that falls from the sky in any form; hail arrives as solid ice pellets and rain as liquid drops. Water vapour is invisible gas in the air, not falling precipitation; the season does not define hail versus rain; and both are measured as a depth in millimetres, not in percent or degrees.

19. (D) Two gauges placed together in one spot should record nearly the same depth from the same storm, so a tenfold gap signals a problem such as a blockage or leak. A storm does not deposit ten times the water a metre apart for no reason, a rain gauge does not measure temperature, and rainfall does not systematically double over short distances.

20. (A) Air has weight, and the pressure at any point is the downward push of the whole column of air above it. Water vapour content is humidity, moving air is wind, and the warmth of high air is temperature—none of these is what pressure measures.

21. (A) Pressure comes from the weight of air above you; the higher you go, the less air is left overhead, so pressure decreases with altitude. Being "nearer the top" actually means less air above, not more; air thins with height rather than staying constant; and pressure drops gradually, never to zero at a mountain pass.

22. (D) Pressure falls with altitude because less air lies above a high place, so a lower reading at the hill station is exactly what is expected—no fault is involved. The difference is due to height, not a broken instrument, the coast's temperature, or rainfall amount.

23. (D) Pressure is set by how much air sits above a point, and high up there is less air overhead, so pressure is lower, not higher—the child has it backwards. Distance to the Sun, the temperature of the ground, and recent rainfall are not what determines the weight of the overlying air.

24. (C) A region of unusually low pressure is a depression, which can intensify into a storm, so a sharply falling barometer warns that rough weather may be coming. Low pressure is linked to unsettled, not clear, weather; the drop is a real reading, not a fault; and a depression is associated with rising cloud and rain rather than safe dry air.

25. (A) Rising pressure is generally linked with sinking, settling air and fair weather, so clearer skies are expected. Falling pressure—not rising—warns of storms and rain, and rising pressure does not make wind cease while the air dampens.

26. (C) Air flows from high pressure to low pressure, and that flow is wind. It does not run from low to high (that is the wrong direction), and although temperature and moisture differences create pressure differences, it is the pressure gradient itself that directly drives the wind.

27. (A) Wind moves from high to low pressure, so with M at 1020 mb and N at 1004 mb the air flows from M to N. It does not flow from the low side to the high side, nor merely circle in place, nor go only straight up when there is a clear horizontal pressure difference.

28. (B) The bigger the pressure difference over a given distance, the harder the air is pushed and the faster the wind, so Pair 2's wind is stronger. A larger difference speeds wind up rather than slowing it, wind speed is not independent of pressure, and the direction still runs high-to-low, not reversed.

29. (D) The wind vane points to show where the wind comes from (direction) and the anemometer's spinning cups measure how fast it blows (speed). Barometers and thermometers measure pressure and temperature, not wind; the wind-vane/anemometer roles must not be swapped; and rain gauges and hygrometers measure rainfall and humidity.

30. (B) The vane swings until its larger tail catches the wind and its arrow points toward the wind's source, giving wind direction. Speed is the anemometer's job, water vapour is humidity (hygrometer), and counting raindrops relates to rainfall (rain gauge).

31. (D) The faster the wind, the faster the anemometer's cups spin, so quick spinning means a high wind speed. Pressure is read on a barometer, humidity on a hygrometer, and direction on a wind vane—none of which is what the spinning cups indicate.

32. (D) A wind sock hangs limp in calm air and stretches out level when the wind is strong, while its open end points the way the wind blows—so a straight, level sock signals a strong wind and gives direction. A level sock is the opposite of calm, and a wind sock indicates neither rainfall nor air pressure.

33. (B) A high relative humidity means the air is nearly saturated, holding close to all the vapour it can, so 95 % describes very damp, sticky air. It is the opposite of dry, does not by itself mean rain is falling, and percent humidity is unrelated to a degree reading of temperature.

34. (B) A reading of 30 % sits in the dry band (20–40 %), so the air is dry and crisp. The damp-and-sticky range is around 60–80 %, rain is not implied by a humidity figure, and 100 % (not 30 %) would mean fully saturated air.

35. (A) Low humidity means the air still has plenty of room to take up water vapour, so moisture leaves the clothes quickly and they dry fast. The day is warm (not too cold), the difference is about how much vapour the air can still accept, not pressure squeezing water in, and humidity does not stop the wind.

36. (D) Rain needs the vapour to condense, which usually requires the air to cool to its saturation point; high humidity makes rain more likely but does not guarantee it. The hygrometer is not necessarily faulty, humidity and rainfall are related (humidity supplies the moisture), and 90 % means very moist, not dry.

37. (A) A barometer reads pressure (in millibars) and a thermometer reads temperature (in degrees); these are distinct weather elements. They do not both measure temperature or humidity, and neither measures rainfall or wind—those belong to the rain gauge and the anemometer/wind vane.

38. (A) Humidity is measured by a hygrometer, not a barometer (which measures pressure), so that pairing is wrong. Temperature/thermometer, rainfall/rain gauge and wind-speed/anemometer are all correctly matched.

39. (B) Atmospheric pressure needs a barometer, which the station lacks, so pressure cannot be recorded. Temperature, rainfall and humidity are covered by the thermometer, rain gauge and hygrometer it does own.

40. (C) Wind direction is read on a wind vane and air moisture on a hygrometer, so that pair fits the task. The anemometer gives speed (not direction) and the barometer gives pressure (not moisture); a rain gauge and thermometer measure rain and temperature; and a barometer measures pressure, not moisture.

41. (C) Nature's clues are local and qualitative, warning only roughly and only for the near future, which is their main limitation. They are not always right, they give no precise numbers, and a single animal's behaviour cannot represent conditions across a whole country.

42. (A) Meteorology is the systematic, scientific study of weather: it uses instruments to take exact readings and combines records from many stations across time, which is what lets it forecast. It does not depend on one animal, it is built on instruments, and gathering data over time is precisely what allows it to predict, not merely report today.

43. (B) An AWS senses and logs readings automatically, so it can keep working where it is hard or risky for people to remain. It does not need a human reader, it is not limited to cities or daylight, and like any weather station it records present weather, not multi-decade climate.

44. (C) The India Meteorological Department is the official agency that collects weather data and issues India's forecasts and warnings. The other names are invented and have no such national role.

45. (A) The Sun's energy heats land, sea and air unevenly, and that uneven heating sets temperature, pressure differences, winds and the water cycle in motion—so the Sun drives the weather. The Moon's pull mainly affects tides, and instruments such as wind vanes and barometers only measure the weather; they do not cause it.

46. (C) Because different surfaces warm at different rates, the air above them differs in temperature and pressure, and air then flows from high to low pressure—that flow is wind. The Sun does not heat all surfaces equally, its rays do not physically push the air along, and the example involves warming and rising air, not freezing.

47. (D) Direct, overhead sunlight in the tropics concentrates its energy on a small area, whereas the same beam hits the poles at a low angle and spreads thinly, so the tropics warm more. Distance to the Sun is essentially the same everywhere, the poles do not get more daily sunshine across the year, and there is no stream of cold space-air singling out the tropics.

48. (D) Temperature climbs while the Sun heats the surface and falls once that heating stops at night, reaching its lowest near dawn—clear evidence the Sun is the driving heat source. Barometers, humidity and wind speed only describe other weather elements; they do not control the day-night temperature cycle.

49. (B) Falling pressure, rising humidity, strengthening wind and thickening dark cloud together point to an approaching storm with rain. These trends are the opposite of clear, calm or unchanging weather, and a cloudless evening contradicts the dark cloud building in the table.

50. (B) A pressure drop signals air moving toward a low—a depression—which can grow into a storm, so it is an early storm warning. Fair weather goes with rising pressure, the station's altitude is fixed and not changing during one afternoon, and falling pressure tends to bring more wind, not its end.

51. (C) Sea-level pressure runs near 1000 mb, a comfortable temperature is a couple of dozen degrees Celsius, and humidity is a percentage that cannot exceed 100, so 1008 mb / 27 °C / 65 % is the only assignment where every value is physically sensible. Each wrong option produces impossible figures, such as humidity above 100 % or a temperature of 1008 °C.

52. (B) Forecasts cannot change the weather, but the advance notice lets people protect lives and property—keeping boats ashore, clearing drains. They do not prevent storms, do not slow how often weather changes, and in fact depend on instruments rather than replacing them.

53. (C) A weather station gathers the full set of instruments in one site and records them at set times so all elements are tracked together. A climate map shows long-term averages, a wind sock only indicates wind, and a depression is a low-pressure region, not a place where instruments are kept.

54. (B) As altitude rises, pressure falls and the air thins, so each breath holds less oxygen, leaving the climber breathless. The air is not heavier or richer in oxygen up high, it is not a vacuum at a mountain pass, and breathlessness here is due to thin air, not rain.

55. (A) Pressure does fall with height, and the cause is precisely that less air remains overhead to push down—so both statements are true and the reason correctly explains the assertion. The other choices wrongly deny that link or falsely call one of the true statements false.

56. (A) Humidity can be very high without any rain, so the assertion is false; the reason is true, because rain does require cooling that condenses the vapour. The true reason actually explains why high humidity by itself is not enough to cause rain, which is why the assertion fails.

57. (B) The two instruments measure different things, so the assertion is false; the reason states their true, distinct jobs—direction for the vane, speed for the anemometer. The reason in fact shows why the assertion is wrong, so any option calling the assertion true is incorrect.

58. (B) Wind actually blows from high pressure to low pressure, so the assertion has the direction reversed and is false; it is true that moving air tends to even out pressure differences. Because the assertion's direction is wrong, every option that treats it as true is incorrect.

59. (D) Climate is built from records over many years, so a single week of weather is far too short to describe it. A week cannot fix a climate, weather and climate are different time-scales, and a rainy week certainly would not show a "dry" climate.

60. (C) Pressure does decrease with height, is read on a barometer in millibars, and a steep fall signals a possible storm—every part is right. The other statements each chain together errors: pressure falling (not rising) with altitude and being read by a barometer; humidity measured by a hygrometer in percent with 100 % meaning saturated, not dry; and rainfall measured by a rain gauge in millimetres with high humidity not guaranteeing rain.